

Education Platform Enterprise System

Overview

This project is a complete enterprise-grade Education Platform Management System designed for:

- Training Institutes
- EdTech Companies
- Coaching Centers
- Placement Institutes
- Online Learning Platforms

The system covers the full lifecycle:

1. Lead generation
2. Student inquiries
3. Demo sessions
4. Course enrollments
5. Batch management
6. Attendance tracking
7. Assessments
8. Mock interviews
9. Placements
10. Analytics & reporting

Architecture Overview

```
graph TD;
    A[Lead Generation] --> B[Inquiry Management];
    B --> C[Demo Programs];
    C --> D[Student Conversion];
    D --> E[Batch Enrollment];
    E --> F[Training & Attendance];
    F --> G[Tests & Assessments];
    G --> H[Mock Interviews];
```

↓
Placement Preparation
↓
Interview Calls
↓
Offers & Hiring
↓
Analytics & Reporting

Technology Stack

Layer	Technology
Database	MySQL
Backend	Python + FastAPI
Frontend	React
ETL	Python + Pandas
Reporting	Power BI
Workflow	Apache Airflow
Containerization	Docker
Version Control	Git + GitHub

Database Schema Overview

The schema contains 27 tables.

1. RBAC MODULE

Purpose

Controls user roles and permissions.

Tables

roles

Stores user roles.

Column	Description
role_id	Unique role ID
role_name	Admin, Trainer, Counselor

users

Stores all application users.

Column	Description
user_id	Unique user ID
role_id	FK to roles
name	User name
email	User email
priority	Priority level

2. COURSE MODULE

courses

Stores available courses.

Column	Description
course_id	Unique course ID
course_name	Python, Power BI, ML
duration	Course duration
fee	Course fees

3. CRM MODULE

Business Flow

Lead → Inquiry → Demo → Converted Student

inquiries

Stores lead inquiries.

Status Values

- new
 - contacted
 - demo
 - converted
 - lost
-

students

Created only after inquiry conversion.

Relationship:

inquiries (1) -----> (1) students

4. DEMO MANAGEMENT MODULE

Purpose

Manages demo programs before enrollment.

Tables

demo_programs

Stores demo batches.

demo_enrollments

Maps inquiries to demo programs.

demo_schedule

Stores demo session timing.

demo_attendance

Tracks attendance.

Demo Workflow

```
graph TD; Inquiry --> Demo_Enrollment[Demo Enrollment]; Demo_Enrollment --> Demo_Sessions[Demo Sessions]; Demo_Sessions --> Attendance; Attendance --> Conversion;
```

Inquiry
↓
Demo Enrollment
↓
Demo Sessions
↓
Attendance
↓
Conversion

5. TRAINING MODULE

Purpose

Manages actual course batches.

batches

Stores training batches.

Modes

- online
 - offline
-

batch_enrollments

Maps students to batches.

batch_schedule

Stores class schedules.

Features

- backup trainer
 - rescheduled classes
 - holidays
-

batch_attendance

Tracks student attendance.

Training Workflow

```
graph TD;
    Student --> Batch_Enrollment[Batch Enrollment];
    Batch_Enrollment --> Class_Schedule[Class Schedule];
    Class_Schedule --> Attendance;
    Attendance --> Completion;
```

6. ROOM & SEAT MANAGEMENT

Purpose

Supports offline classroom management.

rooms

Stores classrooms.

seats

Stores seats inside rooms.

seat_allocations

Maps students to seats.

Offline Batch Flow

```
Room
↓
Seats
↓
Seat Allocation
↓
Student Seating
```

7. TEST & ASSESSMENT MODULE

Purpose

Tracks student learning progress.

tests

Stores tests.

test_results

Stores marks.

mock_interviews

Stores mock interview feedback.

Features

- technical feedback
 - communication feedback
 - scoring
-

8. PLACEMENT MODULE

Purpose

Manages hiring pipeline.

companies

Stores hiring companies.

job_openings

Stores open positions.

interviews

Stores student interview attempts.

Status

- cleared
 - rejected
-

offers

Stores final hiring offers.

Status

- accepted
 - rejected
-

Placement Workflow

```
graph TD;
    Student --> MockInterview[Mock Interview];
    MockInterview --> CompanyOpening[Company Opening];
    CompanyOpening --> Interview[Interview];
    Interview --> Offer[Offer];
    Offer --> Placement[Placement];
```

9. TRAINER MANAGEMENT MODULE

trainer_availability

Stores trainer schedules.

trainer_leaves

Stores leave records.

trainer_metrics

Stores analytics.

Metrics

- total students
 - average attendance
 - conversions
-

10. AUDIT & LOGGING MODULE

audit_logs

Tracks data changes.

Operation Types

- INSERT
 - UPDATE
 - DELETE
-

Entity Relationship Flow

```
roles
  ↓
users
  ↓
trainers / counselors

courses
  ↓
inquiries
  ↓
students
  ↓
batch_enrollments
  ↓
batches
  ↓
batch_attendance
```

↓
placements

CSV Dataset Overview

The dataset contains:

Table	Approx Records
inquiries	5000
batch_enrollments	5000
test_results	5000
demo_enrollments	2500
interviews	4000
offers	2000
users	300
companies	150
courses	10

Foreign Key Relationships

Important Relationships

Parent Table	Child Table
roles	users
courses	inquiries
inquiries	students
users	demo_programs
demo_programs	demo_schedule
students	batch_enrollments
batches	batch_schedule

Parent Table	Child Table
tests	test_results
companies	job_openings
job_openings	interviews
interviews	offers

Data Loading Order

The ETL must load tables in dependency order.

Correct Order

1. roles
2. users
3. courses
4. inquiries
5. students
6. demo_programs
7. demo_enrollments
8. demo_schedule
9. demo_attendance
10. batches
11. batch_enrollments
12. batch_schedule
13. batch_attendance
14. rooms
15. seats
16. seat_allocations
17. tests
18. test_results
19. mock_interviews
20. companies
21. job_openings
22. interviews
23. offers
24. trainer_availability
25. trainer_leaves
26. trainer_metrics
27. audit_logs

ETL Flow

```
graph TD; A[CSV Files] --> B[Pandas DataFrames]; B --> C[Validation]; C --> D[Transformation]; D --> E[MySQL Tables]; E --> F[Analytics Layer];
```

Suggested Power BI Dashboards

CRM Dashboard

- Lead sources
- Conversion rates
- Counselor performance

Training Dashboard

- Attendance
- Batch completion
- Student performance

Placement Dashboard

- Placement rate
- Company hiring trends
- Offer acceptance rate

Trainer Dashboard

- Trainer performance
- Student feedback

- Attendance trends
-

Suggested ETL Enhancements

Production-Level Features

- Logging
 - Retry handling
 - Incremental loading
 - CDC pipelines
 - SCD Type 2
 - Airflow DAGs
 - Error tables
 - Validation framework
 - Data quality checks
 - Duplicate detection
-

Suggested Future Enhancements

AI Features

- Student dropout prediction
 - Lead scoring model
 - Placement probability model
 - AI chatbot support
 - Recommendation engine
-

Cloud Features

- AWS deployment
 - Docker containers
 - Kubernetes
 - CI/CD pipelines
 - Data lake integration
-

Folder Structure

```
project/
|
|— load_data_enterprise.py
|— requirements.txt
|— README.md
|— edu_platform_enterprise_dataset/
|   |— roles.csv
|   |— users.csv
|   |— courses.csv
|   |— inquiries.csv
|   |— students.csv
|   |— demo_programs.csv
|   |— demo_enrollments.csv
|   |— demo_schedule.csv
|   |— demo_attendance.csv
|   |— batches.csv
|   |— batch_enrollments.csv
|   |— batch_schedule.csv
|   |— batch_attendance.csv
|   |— rooms.csv
|   |— seats.csv
|   |— seat_allocations.csv
|   |— tests.csv
|   |— test_results.csv
|   |— mock_interviews.csv
|   |— companies.csv
|   |— job_openings.csv
|   |— interviews.csv
|   |— offers.csv
|   |— trainer_availability.csv
|   |— trainer_leaves.csv
|   |— trainer_metrics.csv
|   |— audit_logs.csv
```

How to Run

Step 1: Install Dependencies

```
pip install pandas sqlalchemy pymysql
```

Step 2: Create Database

```
CREATE DATABASE edu_platform;
```

Step 3: Configure MySQL Credentials

Update inside:

```
DB_USER = "root"  
DB_PASSWORD = "password"
```

Step 4: Run Loader Script

```
python load_data_enterprise.py
```

Validation Queries

```
SELECT COUNT(*) FROM students;  
SELECT COUNT(*) FROM inquiries;  
SELECT COUNT(*) FROM batch_enrollments;  
SELECT COUNT(*) FROM interviews;  
SELECT COUNT(*) FROM offers;
```

Learning Outcomes

This project helps practice:

- SQL
 - Data modeling
 - ETL pipelines
 - Python automation
 - Data warehousing
 - Power BI
 - Airflow
 - Foreign keys
 - Referential integrity
 - Analytics engineering
 - Enterprise architecture
-

End-to-End Business Lifecycle

```
graph TD; A[Lead Generated] --> B[Inquiry Created]; B --> C[Demo Attended]; C --> D[Student Converted]; D --> E[Batch Training]; E --> F[Attendance & Tests]; F --> G[Mock Interviews]; G --> H[Company Interviews]; H --> I[Offer Released]; I --> J[Student Placed];
```